Total Internal Reflection

SYNOPSIS

Run the light the other way — out of glass into air instead of in — and Snell's law gives a refracted angle that is *bigger* than the one you started with. Push far enough and it reaches 90°. Past that there is no refracted ray at all, and the surface behaves like a perfect mirror.

BEFORE YOU START — PREDICT

Light inside glass meets the surface with air above it. As you tilt the ray further from the normal, the refracted ray in the air tilts faster. **What happens when the refracted ray reaches 90°**, and what can possibly happen after that? Write down a guess before you read on.

PERFORMANCE EXPECTATIONS

1. **Explain** why total internal reflection happens only going from a denser medium to a less dense one.
2. **Derive** $\sin\theta_c = n_r/n_i$ by setting $\theta_r = 90°$ in Snell's law.
3. **Calculate** a critical angle from two refractive indices, and say what happens above and below it.
4. **State** the principle of least time and connect it to the bending of a ray at a boundary.

5. Total internal reflection

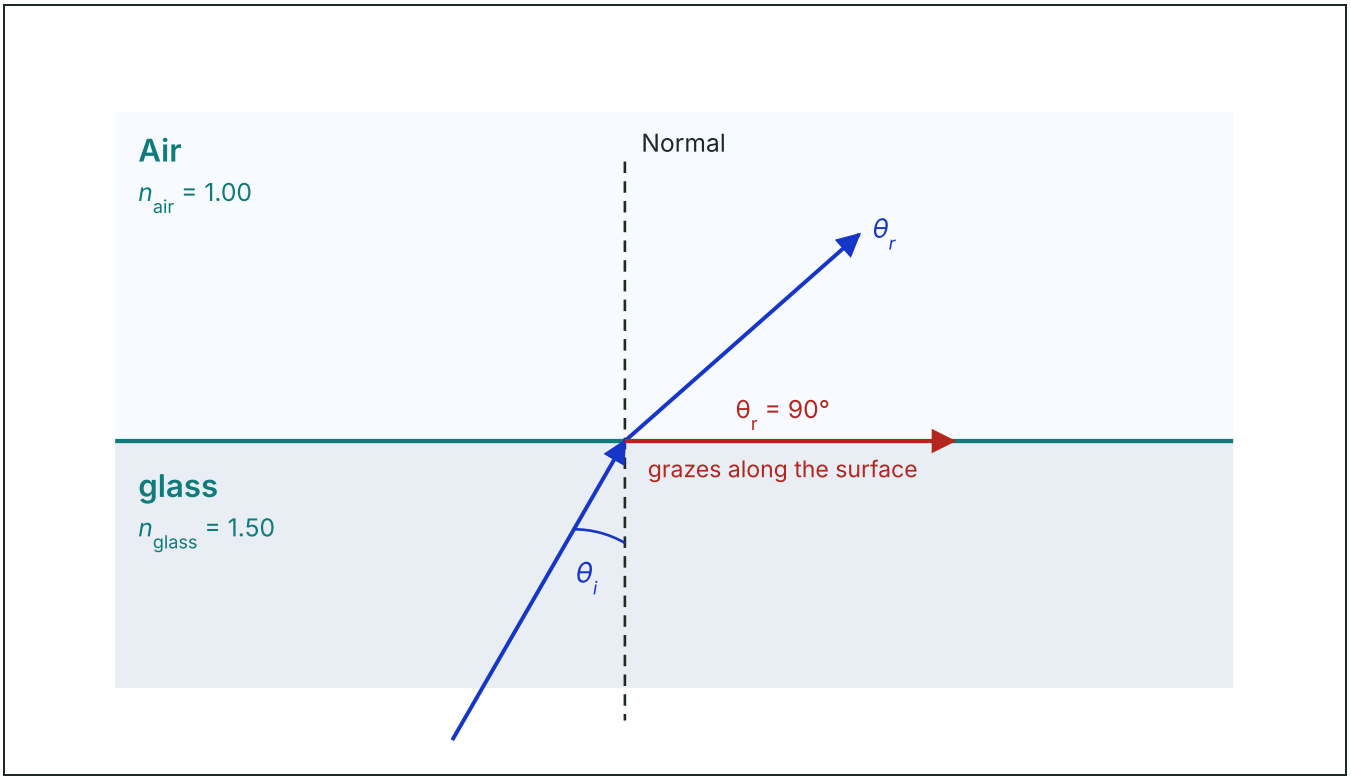


Figure 1. Drawn to scale. The blue ray leaves at θ_i = 30° and refracts to 48.6° — bending *away* from the normal, because it is going into the *less* dense medium. Push θ_i up and θ_r runs ahead of it until it hits 90°. The angle at which that happens is θ_c, and that is the whole of the critical angle.

Your job on Figure 1. The interface and the ordinary refracted ray are drawn for you. Add the ray that arrives at the critical angle, mark θ_c, and show where its refracted ray goes.

1. Find the angle θ_c at which the refracted ray just grazes the surface, in terms of n_i and n_r. Then put the numbers in for glass and air.

THE PUNCHLINE

$$\sin \theta_c = n_r/n_i$$

Below θ_c some light leaves and some reflects. At θ_c the refracted ray lies flat along the surface. Above it, **none** of the light gets out — it is all reflected back inside, which is why the equation is the whole working principle of an optical fibre.

Why light bends at all: the principle of least time

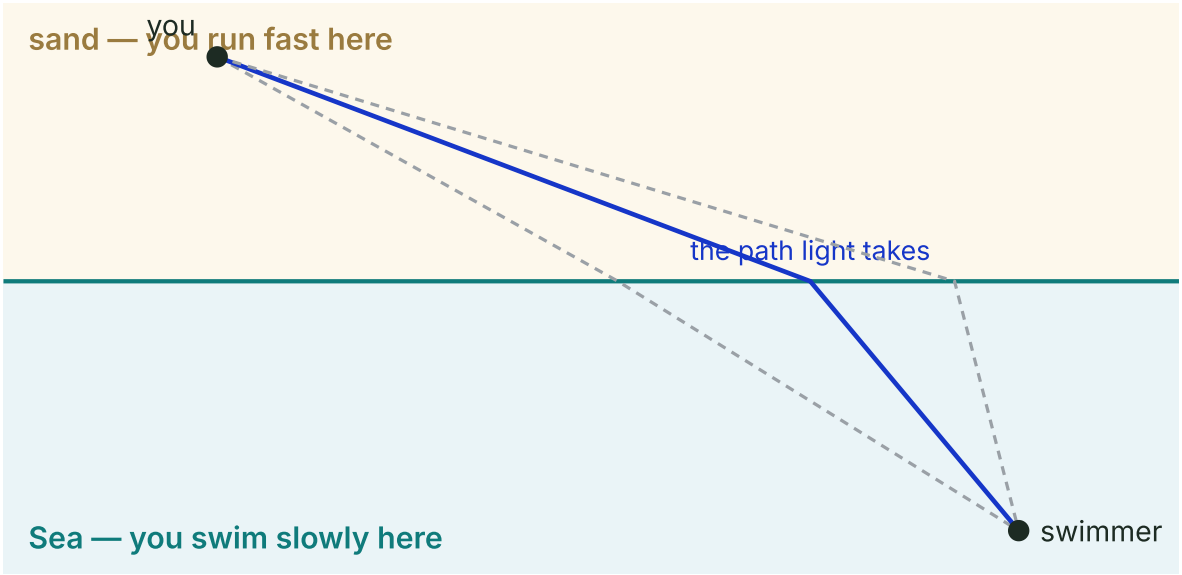


Figure 2 — the principle of least time. Running is fast, swimming is slow, so the quickest route is not the straight line and not the one with the least water in it, but a bent one — more sand, less sea. Light does the same thing at a boundary, and the angle it chooses is the one Snell's law gives.

Looking ahead

That closes the first chapter of this module — what light is, and what a single flat surface does to it. Next the surfaces start to curve, and curved surfaces make **images**.

2. Total internal reflection can happen when light travels…

○ from a less dense medium into a denser one. ○ from a denser medium into a less dense one. ○ in either direction, provided the angle is large enough. ○ only in a vacuum.

3. A ray inside glass ($\theta_c = 41.8°$) meets the surface at 55° from the normal. What happens?

○ It refracts out at 90°, along the surface. ○ It refracts out at some angle less than 90°. ○ None of it leaves: it is totally internally reflected. ○ It is absorbed by the surface.

4. A material has a critical angle of 34.0° against air. What is its refractive index?

Check Clear

(adds about 420 px to the page; re-measure and re-declare the wrapper height) --> Recording to come. Print the guided notes (link just below) and fill them in as it goes.